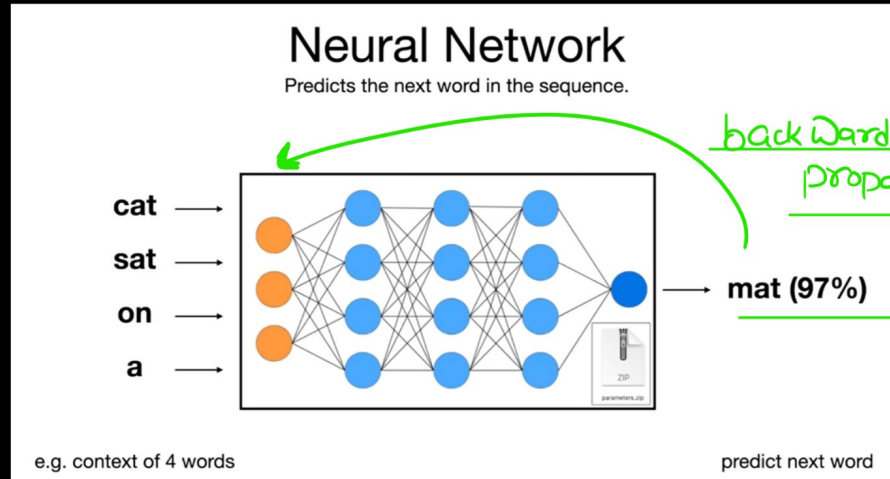


Pretraining

LLM $\rightarrow$ parameters + run.c



① self supervised learning

② Loss function $\uparrow\uparrow$

$\rightarrow$ The sun rises in the

LLM

$\rightarrow$ "West"

③ Probability Engine

$\hat{y} \leftarrow$ prediction

Data $\rightarrow$ by 2026, models are trained on
15 to 30 trillion
tokens

① open web

② digital library

③ Github

④ mathematical proofs

The Speed-Reading Savant Imagine a person who can read 1,000 pages per second. They spend 24 hours a day, 7 days a week, for six months straight, reading every single scrap of text they can find. They don't have a teacher to explain what they are reading; they just observe that the word "Cloud" often appears near "Rain," and "def" in Python is usually followed by a function name.

→ base model (foundational model)

- ① Mastery of Grammar: It knows exactly how to structure a sentence in hundreds of languages.
- ② Approximate World Knowledge: It "knows" who the President is or how a combustion engine works because it has seen those patterns thousands of times.
- ③ Pattern Mimicry: It is an incredible mimic. If you start a Shakespearean play, it will continue in that exact style. If you start a Python script, it will continue writing code.

10 billion parameters

→ terrible coder

70 billion parameters

→ cracking the code &

Can solve any complex
algorithmic problem

"The brilliant but useless" Paradox

What is the capital of India?

What is the capital of Bangladesh?

quiz

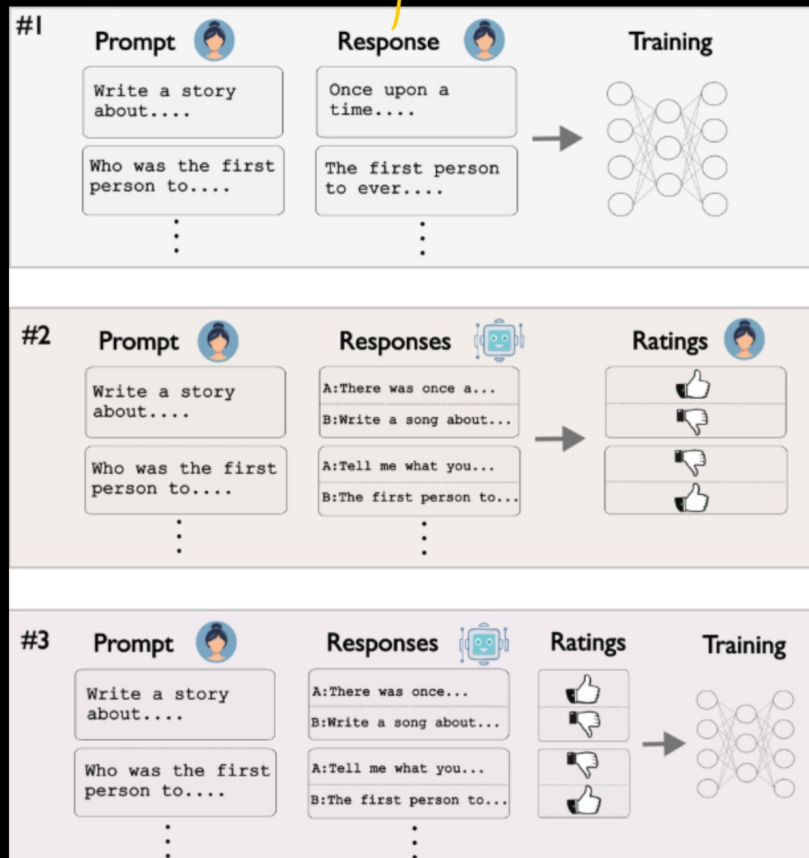
questions

Post-training & fine tuning

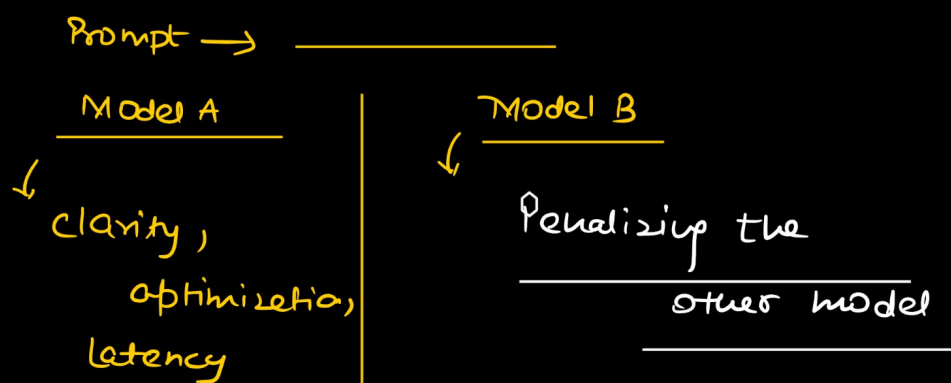
→ ① Supervised fine tuning

Instruction-Response Pairs (SFTs)

user assistant → 1,00,000 files
SFT Training



② RLHF ⇒ Reinforcement Learning with Human feedback



Model A performed better
than Model B

RLHF training

Step 1: Generating the Options The model is given a prompt (e.g., "Tell me a joke about a computer") and generates multiple different responses—Response A, Response B, and Response C.

Step 2: The Human Judge Human reviewers look at these responses and rank them. They don't have to write a new answer; they just indicate their preference (e.g., "Response B is more helpful and less robotic than Response A").

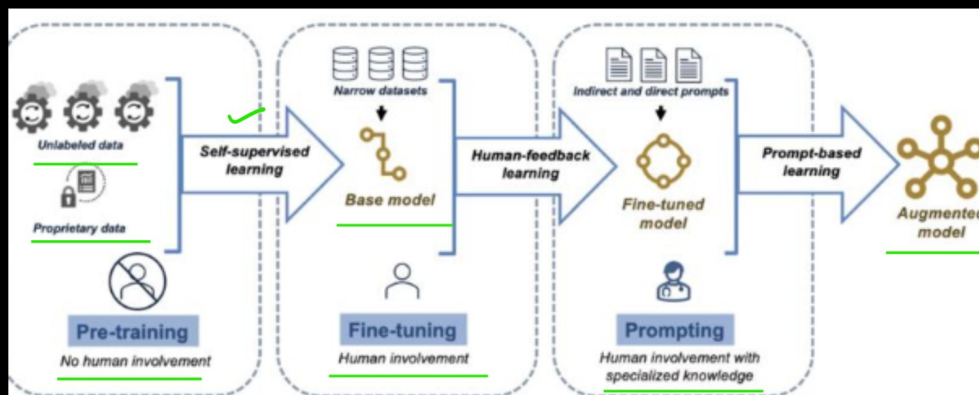
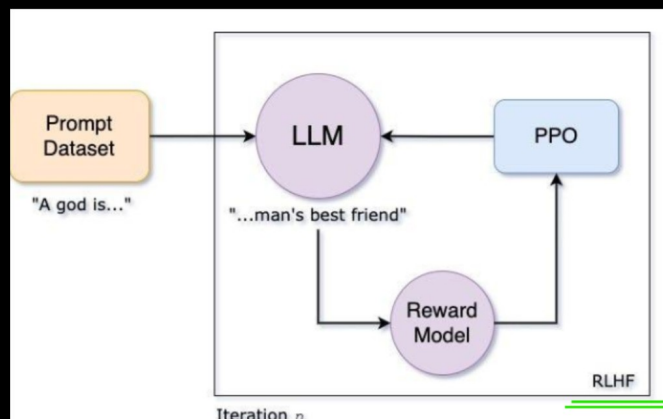
Step 3: Training the "Reward Model" A separate, smaller Transformer is trained specifically to predict these human preferences. This "Reward Model" becomes a digital referee that knows what a "good" human-like response looks like.

Step 4: The Final Update (PPO/DPO) The main LLM is then "played" against the Reward Model. If the LLM generates a response that the Reward Model likes, it receives a mathematical "reward," and its weights are updated to make that behavior more likely in the future.

LLM

training

The Stand-up Comedian Imagine a comedian who has memorized every joke in the library (Pretraining) and knows how to follow a setlist (SFT). RLHF is the process of them performing in front of a live audience. They tell a joke; if the audience laughs (Human Feedback), they keep it in the act. If the audience groans, they change their timing or cut the joke. Over time, the comedian learns not just what to say, but how to say it to get the best reaction.



Summary: how to train your ChatGPT



<USER>
Can you write a short introduction about the relevance of the term "monopsony" in economics? Please use examples related to potential monopsonies in the labour market and the relevant research.

<ASSISTANT>
"Monopsony" refers to a market structure where there is only one buyer for a particular good or service. In economics, this term is particularly relevant in the labour market, where a monopsony employer has significant power over the wages and working conditions of their employees. The presence of a monopsony can result in lower wages and reduced employment opportunities for workers, as the employer has the incentive to increase wages or provide better working conditions...

every
~year

Stage 1: Pretraining

1. Download ~10TB of text.
2. Get a cluster of ~6,000 GPUs.
3. Compress the text into a neural network, pay ~\$2M, wait ~12 days.
4. Obtain base model.

public
datasets

Stage 2: Finetuning

1. Write labeling instructions.
2. Hire people (or use scale.ai!), collect 100K high quality ideal Q&A responses, and/or comparisons.
3. Finetune base model on this data, wait ~1 day.
4. Obtain assistant model.
5. Run a lot of evaluations.
6. Deploy.
7. Monitor, collect misbehaviors, go to step 1.

Turing

google
apple
amazon

SFT,
RLHF

SFT

RLHF

every
~week

OpenAI → add \$5 as credit

Gemini

Google

free tier

personal

project